

The Path To Linux [Volume 1]

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1. Introduction:

1.1. What this book is?

Welcome to **The Linux Path**.

If you're holding this book, chances are you've heard of Linux but aren't quite sure where to begin. Maybe you've used Windows or MacOS your whole life.

Maybe you've heard that Linux is difficult, only for programmers, or that you need to memorize hundreds of commands before you can accomplish anything.

None of that is true.

Linux is simply *another way of using a computer*, one that gives you more control, more freedom, and a deeper understanding of the machine in front of you.

Like any new skill, it can seem unfamiliar at first, but with the right approach it becomes logical and rewarding.

1.2. Who it's for?

This Book was written with one goal in mind: to teach Linux from the ground up. You don't need programming experience. You don't need to know how an operating system works. If you can use a keyboard, follow instructions, and are willing to experiment, you already have everything you need.

1.3. How to use this book?

Throughout this book you'll find practical exercises designed to reinforce what you've learned. Whenever possible, I encourage you to type every command yourself rather than simply reading about it. Linux is learned by using it.

By the end of this volume you will be able to navigate the Linux filesystem with confidence, manage files and directories, understand permissions, edit text files via the terminal, install software, use system documentation, and perform the everyday tasks expected of a Linux user.

This is only the beginning of your journey. The goal of this book is not to make you an expert overnight, but to give you a solid foundation upon which every future Linux skill can be built.

2. What is Linux

2.1. History

-> The Foundation: GNU (1983)

Before Linux existed, Richard Stallman started the **GNU project**. His goal was to create a completely free, Unix-like operating system.

By the early 1990s, the GNU team had build almost all the necessary components of an OS: a compiler, a text editor, and various libraries, but they were missing the most critical part: **the kernel** (we will get to this in the next chapter)

-> The Spark: Linus Torvalds (1991)

In 1991, Linus Torvalds, a 21-year old student at the University of Helsinki in Finland, became frustrated with the operating system he was using for school (Minix). He wanted something more powerful but couldn't afford a commercial Unix License.

As a hobby, Linus began writing his own kernel. On August 25, 1991, he posted a famous message to a computer newsgroup:

I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu)....

-> It's all coming together: GNU/Linux

Linus had the kernel, but he didn't have a full suite of tools (like a shell or a file manager). Meanwhile, the GNU project had the tools, but no kernel.

Linus released his kernel under the GNU General Public License (GPL). This was a pivotal move because it meant anyone could see the code, modify it, and redistribute it, provided they kept the software open source. This encouraged developers worldwide to contribute their own improvements, leading to an explosive growth.

When the GNU tools were combined with the Linux kernel, a complete operating system was born, **GNU/Linux**.

3. The Linux kernel